

Lecture 6

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- Differentiability in $\mathbb{R}^n$ and $\mathbb{R}^{m \times n}$
- Gradient and Jacobian revisited
- Computation of derivatives for linear and quadratic functions
- Product, quotient, and chain rules for multivariate functions

- Differentiability
- Total derivative
- Linear map
- Gradient
- Jacobian
- Directional derivative
- Partial derivative
- Chain rule
- Product rule
- Quotient rule
- Forward propagation
- Backward propagation

The derivative as a linear map

Tangent line vs linear approximation

Draw picture.

The derivative in the univariate case

Let's remember our definition of the derivative in the univariate case:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

In other words,

$$0 = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x) - f'(x)h}{h}$$

$$0 = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x) - f'(x)h}{h}$$

Let's look at that last term: $f'(x)h$.

This is a function of h . We could call it $[Df(x)](h) = f'(x)h$.

In the limit that h is small, the function $[Df(x)](h) = f'(x)h$ is called the *differential* of f at x . But confusingly, the function is also sometimes called the *derivative* of f at x , and that's the language we are going to use here.

So when we say *derivative* in this course, we will often mean this usage, of the differential.

And when we say **the derivative is a linear map**, we are definitely talking about this usage.

Differentiability in $\mathbb{R}^n$ and $\mathbb{R}^{m \times n}$

Let $\mathbb{U}$ and $\mathbb{V}$ be finite-dimensional vector spaces with norms and let Ω be an open subset of $\mathbb{U}$.

Definition (Definition: Derivative of multivariate functions)

A function $f : \Omega \rightarrow \mathbb{V}$ is differentiable at $\mathbf{u} \in \Omega$ if there exists a linear map $Df(\mathbf{u}) : \mathbb{U} \rightarrow \mathbb{V}$ such that

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{\|f(\mathbf{u} + \mathbf{h}) - f(\mathbf{u}) - [Df(\mathbf{u})](\mathbf{h})\|}{\|\mathbf{h}\|} = 0.$$

Equivalently, if we write $\mathbf{h} = \mathbf{v} - \mathbf{u}$, then we get

$$\lim_{\mathbf{v} \rightarrow \mathbf{u}} \frac{\|f(\mathbf{v}) - f(\mathbf{u}) - [Df(\mathbf{u})](\mathbf{v} - \mathbf{u})\|}{\|\mathbf{v} - \mathbf{u}\|} = 0.$$

$Df(\mathbf{u})$ is called the derivative or total derivative of f at $\mathbf{u}$.

The most important thing to keep in mind is that

A DERIVATIVE IS A LINEAR MAP

In particular $Df(\mathbf{u})$ is a *function*.

Like any function, to specify $Df(\mathbf{u})$, we need to specify three things:

1. its domain (i.e., $\mathbb{U}$),
2. its codomain (i.e., $\mathbb{V}$),
3. its rule of assignment (i.e., given an arbitrary vector $\mathbf{h} \in \mathbb{U}$, what is the vector $[Df(\mathbf{u})](\mathbf{h}) \in \mathbb{V}$).

It makes no sense to write $Df(\mathbf{u}) =$ 'some formula in terms of $\mathbf{u}$.' Instead, $Df(\mathbf{u})$ must be written in the way you write down a function:

$$Df(\mathbf{u}) : \mathbb{U} \rightarrow \mathbb{V}, \quad \mathbf{h} \mapsto \text{'some formula in terms of } \mathbf{u} \text{ and } \mathbf{h} \text{'}$$

or

$$Df(\mathbf{u}) : \mathbb{U} \rightarrow \mathbb{V}, \quad [Df(\mathbf{u})](\mathbf{h}) = \text{'some formula in terms of } \mathbf{u} \text{ and } \mathbf{h} \text{'}$$

Note that $[Df(\mathbf{u})](\mathbf{h})$ means the function $Df(\mathbf{u})$ evaluated at $\mathbf{h}$

Some people prefer to write $Df(\mathbf{u}) \cdot \mathbf{h}$, meaning the linear operator $Df(\mathbf{u})$ acting on $\mathbf{h}$.

Since $Df(\mathbf{u}) \in L(\mathbb{U}, \mathbb{V})$ (the space of linear maps from $\mathbb{U}$ to $\mathbb{V}$), we may regard Df as a map $Df : \Omega \rightarrow L(\mathbb{U}, \mathbb{V})$.

The derivative can't depend on the path used to approach the point

If the derivative exists, it must give the same linear approximation regardless of how we approach the point - no matter which direction or along which path.

You're familiar with the univariate case. Consider $y = |x|$.

If we calculate the derivative from the right, we get

$$\lim_{h \rightarrow 0^+} \frac{f(x+h) - f(x)}{h} = 1$$

But from the left, we get

$$\lim_{h \rightarrow 0^-} \frac{f(x+h) - f(x)}{h} = -1$$

Since we get different values, we say that the derivative does not exist. There is no single best linear approximation to the function at the origin.

Now let's think of a function that takes us from $\mathbb{R}^2$ to $\mathbb{R}$:

$$f(\mathbf{x}) : \mathbb{R}^2 \rightarrow \mathbb{R}$$

For this function to be differentiable at a point $\mathbf{x}$, there must exist a unique plane that approximates the surface near that point.

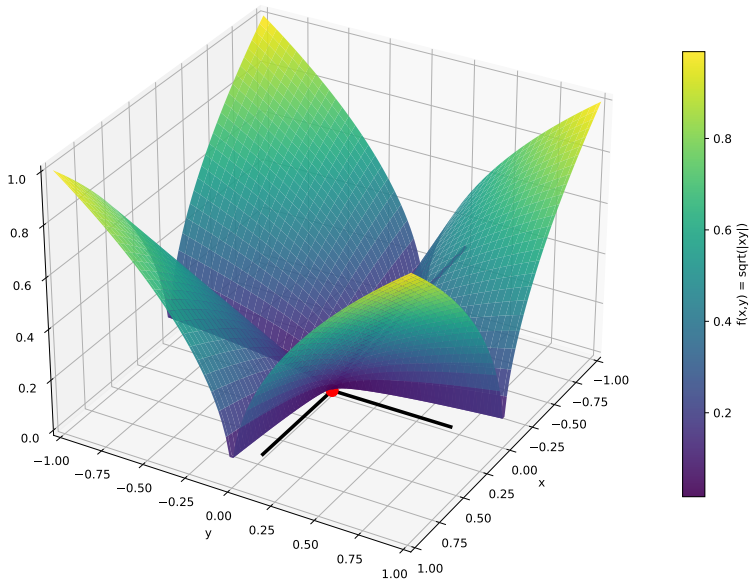
We can find that plane by looking at tangent lines approaching the point from two different directions.

But if we choose a different pair of directions and end up with a different plane, the function cannot be differentiable at that point - even if all partial derivatives exist.

Let's look at an example.

This is the function $f(x, y) = \sqrt{|xy|}$:

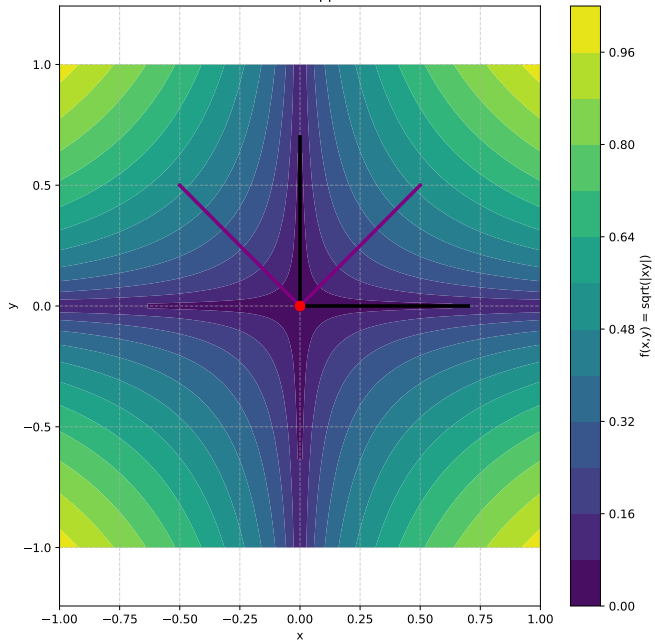
3D Surface Plot: $f(x, y) = \sqrt{|xy|}$



If we approximate this shape by a plane at the origin, using the tangent lines along the x - and y -axes, it would just be a flat plane.

But now imagine turning 45 degrees. You get an entirely different plane!

Contour Plot with Approach Lines



Two points to take home:

- For some functions, the partial derivatives exist but the derivative itself does not.
- What the derivative is doing is finding the best linear approximation to the function near each point.

Gradient revisited

In this course, $\mathcal{U}$ and $\mathcal{V}$ will always be $\mathbb{R}^n$ or $\mathbb{R}^{m \times n}$ or their subspaces (e.g., $\mathbb{S}^n$) or products of these (e.g., $\mathbb{S}^n \times \mathbb{R}^n$) and so the abstract vectors $\mathbf{u}, \mathbf{v}, \mathbf{h}$ above will usually be vectors or matrices.

The most important special case for optimization is when the codomain of the linear map is the field of scalars, $\mathcal{V} = \mathbb{R}$.

The **Riesz representation theorem** tells us that there is a special way to represent any linear map $T : \mathcal{U} \rightarrow \mathbb{R}$: as an inner product with some fixed element in $\mathcal{U}$.

That is, there exists some element $\mathbf{a} \in \mathcal{U}$ such that

$$T(\mathbf{h}) = \langle \mathbf{a}, \mathbf{h} \rangle$$

for all $\mathbf{h} \in \mathcal{U}$.

We'll review this more in the next lecture, but we'll use it now!

The gradient of a function of a vector

Let's consider a function that takes us from $\mathbb{R}^n$ to $\mathbb{R}$.

$$f : \Omega \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$$

Its derivative at the point $\mathbf{x} \in \Omega$ must then be a linear map from $\mathbb{R}^n$ to $\mathbb{R}$.

The Riesz representation theorem tells us that there exists some fixed vector $\mathbf{a}_{\mathbf{x}} \in \mathbb{R}^n$ such that

$$Df(\mathbf{x})(\mathbf{h}) = \langle \mathbf{a}_{\mathbf{x}}, \mathbf{h} \rangle = \mathbf{a}_{\mathbf{x}}^{\top} \mathbf{h}$$

To be clear what I mean:

There is one vector $\mathbf{a}_{\mathbf{x}}$ for every point $\mathbf{x}$. We can use that $\mathbf{a}_{\mathbf{x}}$ to find the derivative of f near $\mathbf{x}$ at any direction $\mathbf{h} \in \mathbb{R}^n$.

$$Df(\mathbf{x})(\mathbf{h}) = \mathbf{a}_{\mathbf{x}}^\top \mathbf{h}$$

What is this special vector $\mathbf{a}_{\mathbf{x}}$?

It is exactly the gradient of f at $\mathbf{x}$.

$$\nabla f(\mathbf{x}) = \mathbf{a}_{\mathbf{x}}$$

(You can forget about $\mathbf{a}_{\mathbf{x}}$ now... just use $\nabla f(\mathbf{x})$ instead.)

The gradient of $f : \Omega \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ may be *defined* to be the vector $\nabla f(\mathbf{x}) \in \mathbb{R}^n$ such that

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{|f(\mathbf{x} + \mathbf{h}) - f(\mathbf{x}) - \nabla f(\mathbf{x})^\top \mathbf{h}|}{\|\mathbf{h}\|} = 0$$

This is because if the expression above is true, then $\nabla f(\mathbf{x})^\top \mathbf{h}$ matches the definition of the derivative we showed earlier in the lecture:

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{\|f(\mathbf{x} + \mathbf{h}) - f(\mathbf{x}) - [Df(\mathbf{x})](\mathbf{h})\|}{\|\mathbf{h}\|} = 0 \tag{1}$$

Let's look at a parabola: $f(\mathbf{x}) = x_1^2 + x_2^2$.

The gradient at the point x_1, x_2 is $\nabla f(\mathbf{x}) = [2x_1, 2x_2]$.

Then the derivative at that point is the linear map given by

$$Df(\mathbf{x})(\mathbf{h}) = \nabla f(\mathbf{x})^\top \mathbf{h} = 2x_1 h_1 + 2x_2 h_2$$

So to find the linear approximation to f at the point $\mathbf{x} + \mathbf{h}$, we can just use the gradient!

$$f(\mathbf{x} + \mathbf{h}) \approx f(\mathbf{x}) + \nabla f(\mathbf{x})^\top \mathbf{h}$$

Again: what exactly do we mean by “derivative”?

When we say “derivative” in this context, we mean specifically the linear map $Df(\mathbf{x})$, not the differential form.

In our multivariable setting: - The derivative $Df(\mathbf{x})$ is a linear map from $\mathbb{U}$ to $\mathbb{V}$ - This map can be applied to vectors: $Df(\mathbf{x})(\mathbf{h})$ - The linear approximation of the function is $f(\mathbf{x}) + Df(\mathbf{x})(\mathbf{h})$

So the derivative is not the coefficient of $d\mathbf{x}$ - it's the entire linear map itself.

For functions $f : \mathbb{R}^n \rightarrow \mathbb{R}$, we can represent this map using the gradient:

$$Df(\mathbf{x})(\mathbf{h}) = \nabla f(\mathbf{x})^T \mathbf{h}$$

The gradient of a function of a matrix

Let's do the same thing for a function that takes us from $\mathbb{R}^{m \times n}$ to $\mathbb{R}$.

$$f : \Omega \subseteq \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$$

The Riesz representation theorem tells us that there exists some fixed matrix $A_X \in \mathbb{R}^{m \times n}$ such that

$$Df(X)(H) = \langle A_X, H \rangle = \text{tr}(A_X^\top H)$$

The gradient of $f : \Omega \subseteq \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ may be defined to be the matrix $\nabla f(X) \in \mathbb{R}^{m \times n}$ such that

$$\lim_{H \rightarrow 0} \frac{|f(X + H) - f(X) - \text{tr}(\nabla f(X)^\top H)|}{\|H\|} = 0. \quad (2)$$

You can show that formulas of $\nabla f(\mathbf{x})$ and $\nabla f(X)$ in terms of partial derivatives satisfy these two definitions respectively

but the more important thing to realize is that Equation 1 and Equation 2 give you a way to find gradients without writing down a single partial derivative!

The derivative in Equation 1 is

$$Df(\mathbf{x}) : \mathbb{R}^n \rightarrow \mathbb{R}, \quad [Df(\mathbf{x})](\mathbf{h}) = \nabla f(\mathbf{x})^\top \mathbf{h} \quad (3)$$

while the derivative in Equation 2 is

$$Df(X) : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}, \quad [Df(X)](H) = \text{tr}(\nabla f(X)^\top H) \quad (4)$$

and Equation 3 and Equation 4 may be taken to be the definitions of $\nabla f(\mathbf{x})$ and $\nabla f(X)$ respectively.

Generalizing the definition of the gradient

In general for *any* vector space $\mathbb{U}$ with inner product $\langle \cdot, \cdot \rangle$, the gradient of a function $f : \Omega \subseteq \mathbb{U} \rightarrow \mathbb{R}$ is an element $\nabla f(\mathbf{u}) \in \mathbb{U}$ such that

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{|f(\mathbf{u} + \mathbf{h}) - f(\mathbf{u}) - \langle \nabla f(\mathbf{u}), \mathbf{h} \rangle|}{\|\mathbf{h}\|} = 0, \quad (5)$$

or, in other words,

$$Df(\mathbf{u}) : \mathbb{U} \rightarrow \mathbb{R}, \quad [Df(\mathbf{u})](\mathbf{h}) = \langle \nabla f(\mathbf{x}), \mathbf{h} \rangle.$$

Equation 1 and Equation 2 are just special cases of Equation 5 with inner products $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top \mathbf{y}$ and $\langle X, Y \rangle = \text{tr}(X^\top Y)$ respectively.

We sometimes drop absolute values in the numerator in Equation 5, ditto for Equation 1 and Equation 2, since $\lim_{\mathbf{y} \rightarrow \mathbf{x}} |f(\mathbf{y})| = 0$ iff $\lim_{\mathbf{y} \rightarrow \mathbf{x}} f(\mathbf{y}) = 0$.

Exercise

Let $\Omega \subseteq \mathbb{S}^n \times \mathbb{R}^n$ and $f : \Omega \rightarrow \mathbb{R}$. What should the gradient of f be?

Jacobian revisited

For a vector-valued function $f : \Omega \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$, its Jacobian may be defined to be the matrix $Jf(\mathbf{x}) \in \mathbb{R}^{m \times n}$ such that

$$\lim_{\mathbf{h} \rightarrow 0} \frac{\|f(\mathbf{x} + \mathbf{h}) - f(\mathbf{x}) - Jf(\mathbf{x}) \cdot \mathbf{h}\|}{\|\mathbf{h}\|} = 0 \quad (6)$$

where $\cdot$ is matrix-vector product. This is a generalization of Equation 1: when $m = 1$, we get

$$Jf(\mathbf{x}) = \nabla f(\mathbf{x})^\top$$

The derivative in Equation 6 is

$$Df(\mathbf{x}) : \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad [Df(\mathbf{x})](\mathbf{h}) = Jf(\mathbf{x}) \cdot \mathbf{h} \quad (7)$$

and this may be taken to be the definition of $Jf(\mathbf{x})$.

Simple calculations: linear and quadratic

How do we actually compute the derivatives, without using partial derivatives?

The trick is to just expand $f(\mathbf{x} + \mathbf{h}) - f(\mathbf{x})$ or $f(X + H) - f(X)$ and look out for the linear terms where $\mathbf{h}$ or H appears exactly once

These will ultimately give you the formula for your $[Df(\mathbf{x})](\mathbf{h})$ or $[Df(X)](H)$.

The derivative of a linear map is itself.

Huh? Let's pause here for a second.

We are thinking of the *derivative* as a linear approximation to a function. So if the function itself is linear, then it makes sense that the derivative will match the function itself.

Let's make this more precise.

Let $f : \mathbb{U} \rightarrow \mathbb{V}$ be a linear map, i.e., $f(a_1 \mathbf{u}_1 + a_2 \mathbf{u}_2) = a_1 f(\mathbf{u}_1) + a_2 f(\mathbf{u}_2)$ for all $a_1, a_2 \in \mathbb{R}, \mathbf{u}_1, \mathbf{u}_2 \in \mathbb{U}$.

We claim that for any $\mathbf{u} \in \mathbb{U}$, f is differentiable, with $Df(\mathbf{u}) = f$. This is easy:

$$\begin{aligned} \lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{\|f(\mathbf{u} + \mathbf{h}) - f(\mathbf{u}) - [Df(\mathbf{u})](\mathbf{h})\|}{\|\mathbf{h}\|} &= \lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{\|f(\mathbf{u}) + f(\mathbf{h}) - f(\mathbf{u}) - [Df(\mathbf{u})](\mathbf{h})\|}{\|\mathbf{h}\|} \\ &= \lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{\|f(\mathbf{h}) - [Df(\mathbf{u})](\mathbf{h})\|}{\|\mathbf{h}\|} = 0 \end{aligned}$$

We see that the above is true if we set $Df(\mathbf{u}) = f$ in the last step.

Note that $Df(\mathbf{u})$ is independent of $\mathbf{u}$. This is not the case in general but is a special feature of linear maps.

Concrete examples of derivatives of linear maps

A simple example of a linear univariate function

First I want to look at a very simple example of a linear function in $\mathbb{R}$: $f(x) = 3x$.

We know that $f'(x) = 3$.

Suppose we want to approximate $f(x + h)$ for a small h , say near $x = 1$.

The exact expression is $f(x + h) = 3(x + h) = 3x + 3h$.

The approximation is given by

$$f(x + h) \approx f(x) + f'(x)h = 3x + 3h$$

The approximation equals the function, because we are using the derivative to approximate the function as a straight line – and it actually **is** a straight line.

Now let's look at a linear function in $\mathbb{R}^2$. Let's pick $f(\mathbf{x}) = 3x_1 + 4x_2$.

Since this is a linear map, we know from above that $Df(\mathbf{x}) = f$, i.e., $Df(\mathbf{x}) = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$.

So the derivative of f in a direction $\mathbf{h}$ near $\mathbf{x}$ is given by

$$Df(\mathbf{x})^\top \mathbf{h} = \begin{bmatrix} 3 & 4 \end{bmatrix} \cdot \begin{bmatrix} h_1 \\ h_2 \end{bmatrix} = 3h_1 + 4h_2.$$

Let's use this derivative to approximate $f(\mathbf{x} + \mathbf{h})$ for a small $\mathbf{h}$.

The approximation is given by

$$f(\mathbf{x} + \mathbf{h}) \approx f(\mathbf{x}) + Df(\mathbf{x}) \cdot \mathbf{h} = 3x_1 + 4x_2 + 3h_1 + 4h_2$$

This is exactly equal to the exact expression for $f(\mathbf{x} + \mathbf{h})$:

$$f(\mathbf{x} + \mathbf{h}) = 3(x_1 + h_1) + 4(x_2 + h_2) = 3x_1 + 4x_2 + 3h_1 + 4h_2$$

Why? The same reason as in the univariate case: f is a linear map, so the derivative is able to perfectly represent it.

Example (Matrix-vector product)

Consider the function that takes a vector and returns the product of that vector with a fixed matrix:

$f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ defined by $\mathbf{u} \mapsto A\mathbf{u}$ for some fixed $A \in \mathbb{R}^{m \times n}$.

This is a linear map and so $Df(\mathbf{x}) = f$ for all $\mathbf{x} \in \mathbb{R}^n$

In other words for any $\mathbf{x} \in \mathbb{R}^n$, $Df(\mathbf{x})$ is the linear map

$$Df(\mathbf{x}) : \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad \mathbf{h} \mapsto A\mathbf{h}.$$

Whenever we can express the derivative in this form, by Equation 7, we must have Jacobian

$$Jf(\mathbf{x}) = A$$

Exercise

Let $\mathbf{a} \in \mathbb{R}^m$ be fixed and constant. Find the derivative of $f : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}^n, X \mapsto X\mathbf{a}$.

Note that even for this simple function, gradient and Jacobian are undefined - it makes no sense to write $\nabla f(X)$ (because f is vector-valued) or $Jf(X)$ (because f is matrix-variate) - which goes to show the limitation of gradient and Jacobian. $Df(X)$ on the other is always defined.

Example (Trace inner product)

Let $A \in \mathbb{R}^{m \times n}$ be fixed and constant.

The derivative of $f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}, X \mapsto \text{tr}(A^\top X)$ is itself since it is linear, i.e.,

$$Df(X) : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}, \quad H \mapsto \text{tr}(A^\top H).$$

Whenever we can express the derivative in this form, by Equation 4, we must have gradient

$$\nabla f(X) = A$$

Exercise

Let $\mathbf{a} \in \mathbb{R}^m$, $\mathbf{b} \in \mathbb{R}^n$ be fixed and constant. Find the derivative of $f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}, X \mapsto \mathbf{a}^\top X \mathbf{b}$ and its gradient at $X \in \mathbb{R}^{m \times n}$.

Quadratic functions

Now we move on to the quadratic case.

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}, \mathbf{x} \mapsto \mathbf{x}^\top A \mathbf{x}$, where $A \in \mathbb{R}^{n \times n}$ is a symmetric matrix.

Note that since

$$A^\top = A, \mathbf{x}^\top A \mathbf{y} = (\mathbf{x}^\top A \mathbf{y})^\top = \mathbf{y}^\top A^\top \mathbf{x} = \mathbf{y}^\top A \mathbf{x}$$

We claim that f is differentiable and that the derivative is

$$Df(\mathbf{x}) : \mathbb{R}^n \rightarrow \mathbb{R}, \mathbf{h} \mapsto 2\mathbf{x}^\top A \mathbf{h}.$$

First step: plug in the function definition.

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{|f(\mathbf{x} + \mathbf{h}) - f(\mathbf{x}) - [Df(\mathbf{x})](\mathbf{h})|}{\|\mathbf{h}\|} = \lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{|(\mathbf{x} + \mathbf{h})^\top A(\mathbf{x} + \mathbf{h}) - \mathbf{x}^\top A\mathbf{x} - [Df(\mathbf{x})](\mathbf{h})|}{\|\mathbf{h}\|}$$

Now we expand the expression inside the limit.

$$= \lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{|\mathbf{x}^\top A\mathbf{x} + \mathbf{h}^\top A\mathbf{x} + \mathbf{x}^\top A\mathbf{h} + \mathbf{h}^\top A\mathbf{h} - \mathbf{x}^\top A\mathbf{x} - [Df(\mathbf{x})](\mathbf{h})|}{\|\mathbf{h}\|}$$

Any term here that is higher than first order in $\mathbf{h}$ will vanish in the limit, so I can cross them out!

Also, because A is symmetric, $\mathbf{x}^\top A\mathbf{h} = \mathbf{h}^\top A\mathbf{x}$, so we can simplify to

$$= \lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{|2\mathbf{x}^\top A\mathbf{h} + \mathbf{h}^\top A\mathbf{h} - [Df(\mathbf{x})](\mathbf{h})|}{\|\mathbf{h}\|}$$

Now we can guess the derivative.

$$[Df(\mathbf{x})](\mathbf{h}) = 2\mathbf{x}^\top A\mathbf{h}$$

Now we plug this in and see that it works.

$$\lim_{\mathbf{h} \rightarrow 0} \frac{|2\mathbf{x}^\top A\mathbf{h} + \mathbf{h}^\top A\mathbf{h} - [Df(\mathbf{x})](\mathbf{h})|}{\|\mathbf{h}\|} = \lim_{\mathbf{h} \rightarrow 0} \frac{|2\mathbf{x}^\top A\mathbf{h} + \mathbf{h}^\top A\mathbf{h} - 2\mathbf{x}^\top A\mathbf{h}|}{\|\mathbf{h}\|} = \lim_{\mathbf{h} \rightarrow 0} \frac{|\mathbf{h}^\top A\mathbf{h}|}{\|\mathbf{h}\|}$$

By Cauchy-Schwarz,

$$\frac{|\mathbf{h}^\top A\mathbf{h}|}{\|\mathbf{h}\|} \leq \frac{\|\mathbf{h}\| \|A\mathbf{h}\|}{\|\mathbf{h}\|} = \|A\mathbf{h}\| \rightarrow 0$$

It works! The limit is 0, so we have found the derivative!

Writing

$$[Df(\mathbf{x})](\mathbf{h}) = 2\mathbf{x}^\top A\mathbf{h} = (2A\mathbf{x})^\top \mathbf{h} = \nabla f(\mathbf{x})^\top \mathbf{h}$$

we see by Equation 3 that the gradient of f at $\mathbf{x} \in \mathbb{R}^n$ is given by

$$\nabla f(\mathbf{x}) = 2A\mathbf{x}$$

Remember, we started with the function $f(\mathbf{x}) = \mathbf{x}^\top A\mathbf{x}$.

So our result here is consistent with the univariate case, $f(x) = ax^2$ has derivative $f'(x) = 2ax$.

Product, quotient, and chain rules

Just as in univariate calculus, complicated functions are built out of simpler functions.

The rules in this section will apply to arbitrary vector spaces $\mathbb{U}, \mathbb{V}, \mathbb{W}$, but as usual we'll be interested in $\mathbb{R}^n$ or $\mathbb{R}^{m \times n}$ or $\mathbb{S}^n$, or their product spaces.

All following results may be restricted to functions defined only on a subset $\Omega \subseteq \mathbb{U}$ instead of the whole space $\mathbb{U}$.

Proposition (Linear combination rule)

Let $f, g : \mathbb{U} \rightarrow \mathbb{V}$ be differentiable at $u \in \mathbb{U}$. Then $\alpha f + \beta g : \mathbb{U} \rightarrow \mathbb{V}$ is also differentiable at $u \in \mathbb{U}$, and

$$D(\alpha f + \beta g)(u) = \alpha Df(u) + \beta Dg(u)$$

for any $\alpha, \beta \in \mathbb{R}$.

Proposition (Chain rule)

Let $g : \mathbb{U} \rightarrow \mathbb{V}$ be differentiable at $u \in \mathbb{U}$ and $f : \mathbb{V} \rightarrow \mathbb{W}$ be differentiable at $g(u) \in \mathbb{V}$. Then their composition $f \circ g : \mathbb{U} \rightarrow \mathbb{W}$ defined by $f \circ g(u) = f(g(u))$ is differentiable at $u \in \mathbb{U}$, and

$$D(f \circ g)(u) = Df(g(u)) \circ Dg(u)$$

Proposition (Product and quotient rules)

Let $f : \mathbb{U} \rightarrow \mathbb{V}$ and $g : \mathbb{U} \rightarrow \mathbb{R}$ be differentiable. Consider the functions

$$\begin{aligned}f \cdot g : \mathbb{U} &\rightarrow \mathbb{V}, & (f \cdot g)(u) &= f(u) \cdot g(u), \\f/g : \mathbb{U} &\rightarrow \mathbb{V}, & (f/g)(u) &= f(u)/g(u),\end{aligned}$$

where $\cdot$ denotes scalar product and f/g is only defined at $u \in \mathbb{U}$ where $g(u) \neq 0$. Then

$$\begin{aligned}[D(f \cdot g)(u)](h) &= f(u) \cdot [Dg(u)](h) + g(u) \cdot [Df(u)](h), \\ \left[D \left(\frac{f}{g} \right) (u) \right] (h) &= \frac{g(u) \cdot [Df(u)](h) - f(u) \cdot [Dg(u)](h)}{g(u)^2}\end{aligned}$$

Again the latter is only defined when $g(u) \neq 0$.

Proof of product and quotient rules

Proof. The same standard proof for product rule, i.e., adding and subtracting terms, works:

$$\begin{aligned} & f(u+h) \cdot g(u+h) - f(u) \cdot g(u) - f(u) \cdot [Dg(u)](h) - g(u) \cdot [Df(u)](h) \\ &= f(u+h) \cdot g(u+h) - f(u+h) \cdot g(u) + f(u+h) \cdot g(u) - f(u) \cdot g(u) \\ &\quad - f(u) \cdot [Dg(u)](h) - g(u) \cdot [Df(u)](h) \\ &= f(u+h) \cdot g(u+h) - f(u+h) \cdot g(u) - f(u) \cdot [Dg(u)](h) \\ &\quad + f(u+h) \cdot g(u) - f(u) \cdot g(u) - g(u) \cdot [Df(u)](h) \\ &= f(u+h) \cdot g(u+h) - f(u+h) \cdot g(u) - f(u+h) \cdot [Dg(u)](h) \\ &\quad f(u+h) \cdot [Dg(u)](h) - f(u) \cdot [Dg(u)](h) \\ &\quad f(u+h) \cdot g(u) - f(u) \cdot g(u) - g(u) \cdot [Df(u)](h) \\ &= f(u+h) \cdot \{g(u+h) - g(u) - [Dg(u)](h)\} \\ &\quad + (f(u+h) - f(u) - [Df(u)](h)) \cdot [Dg(u)](h) + [Df(u)](h) \cdot [Dg(u)](h) \\ &\quad + g(u) \cdot \{f(u+h) - f(u) - [Df(u)](h)\}, \end{aligned}$$

dividing by $\|h\|$ and taking limits $h \rightarrow 0$ gives

$$\begin{aligned} f(u+h) \cdot \left[\frac{g(u+h) - g(u) - [Dg(u)](h)}{\|h\|} \right] &+ \left[\frac{f(u+h) - f(u) - [Df(u)](h)}{\|h\|} \right] \cdot [Dg(u)](h) \\ &+ \frac{[Df(u)](h) \cdot [Dg(u)](h)}{\|h\|} + g(u) \cdot \left[\frac{f(u+h) - f(u) - [Df(u)](h)}{\|h\|} \right] \rightarrow 0. \end{aligned}$$

The reason

$$\frac{[Df(u)](h) \cdot [Dg(u)](h)}{\|h\|} \rightarrow 0$$

is because

$$\|[Df(u)](h)\| \leq \|Df(u)\| \|h\|, \quad \|[Dg(u)](h)\| \leq \|Dg(u)\| \|h\|,$$

where for any linear map $T : \mathcal{U} \rightarrow \mathcal{V}$, we define $\|T\| = \sup_{h \neq 0} \|T(h)\|/\|h\|$ and this is always finite on finite-dimensional spaces.

For quotient rule, just apply product rule to $f \cdot (1/g)$.

Proposition (Composition with scalar-valued functions)

Let $g : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ be differentiable at $X \in \mathbb{R}^{m \times n}$ and $f : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable at $g(X) \in \mathbb{R}$. Then $h : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ defined by $h(X) = f(g(X))$ is differentiable at X , and

$$\nabla h(X) = f'(g(X)) \nabla g(X).$$

As usual, this specializes to $\mathbb{R}^m = \mathbb{R}^{m \times 1}$.

Proposition (Composition with vector-valued functions)

Let $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be differentiable at $\mathbf{x} \in \mathbb{R}^n$ and $f : \mathbb{R}^m \rightarrow \mathbb{R}^p$ be differentiable at $g(\mathbf{x}) \in \mathbb{R}^m$. Then $h : \mathbb{R}^n \rightarrow \mathbb{R}^p$ be defined by $h(\mathbf{x}) = f(g(\mathbf{x}))$ is differentiable at $\mathbf{x}$, and

$$Jh(\mathbf{x}) = Jf(g(\mathbf{x})) \cdot Jg(\mathbf{x})$$

where $\cdot$ is matrix multiplication.

Proposition (Composition with affine functions)

Let $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $h : \mathbb{R}^p \rightarrow \mathbb{R}^m$ defined by $h(\mathbf{x}) = g(A\mathbf{x} + \mathbf{b})$ where $A \in \mathbb{R}^{n \times p}$ and $\mathbf{b} \in \mathbb{R}^n$

$$Jh(\mathbf{x}) = Jg(A\mathbf{x} + \mathbf{b})A$$

The special case where $m = 1$ gives:

$$\nabla h(\mathbf{x}) = A^\top \nabla g(A\mathbf{x} + \mathbf{b}).$$

Corollary (One more product rule)

Let $f, g : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ be differentiable. Then

$$\begin{aligned}\nabla(fg) &= f\nabla g + g\nabla f, \\ \nabla\left(\frac{f}{g}\right) &= \frac{g\nabla f - f\nabla g}{g^2}.\end{aligned}$$

Let $g : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ and $f : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable. Then

$$\nabla(f \circ g) = (f' \circ g) \nabla g$$

Products of many matrices

Repeated composition of functions

If we have a repeated composition of functions like $f = f_k \circ f_{k-1} \circ \cdots \circ f_2 \circ f_1$,

$$\mathbb{R}^{n_0} \xrightarrow{f_1} \mathbb{R}^{n_1} \xrightarrow{f_2} \mathbb{R}^{n_2} \xrightarrow{f_3} \mathbb{R}^{n_3} \xrightarrow{f_4} \cdots \xrightarrow{f_{k-1}} \mathbb{R}^{n_{k-1}} \xrightarrow{f_k} \mathbb{R}^{n_k},$$

then applying Proposition 8.5 multiple times gives us

$$\underbrace{Jf(\mathbf{x})}_{n_k \times n_0} = \underbrace{Jf_k(f_{k-1} \circ \cdots \circ f_1(\mathbf{x}))}_{n_k \times n_{k-1}} \cdot \underbrace{Jf_{k-1}(f_{k-2} \circ \cdots \circ f_1(\mathbf{x}))}_{n_{k-1} \times n_{k-2}} \cdots \underbrace{Jf_2(f_1(\mathbf{x}))}_{n_2 \times n_1} \cdot \underbrace{Jf_1(\mathbf{x})}_{n_1 \times n_0}$$

where again $\circ$ denotes function composition and $\cdot$ denotes matrix product.

Product of k matrices

There are two obvious ways to compute this product of k matrices:

- compute from left to right;
- compute from right to left.

It makes sense to start at whichever end of the chain has the simplest/smallest matrices. That just makes computation easier.

An analogy: imagine you are multiplying

$$4 \times 3 \times 10 \times \sqrt{37}$$

If you multiply from left to right, you can do the first few steps in your head:

$$4 \times 3 \times 10 \times \sqrt{37} = 120 \times \sqrt{37}$$

If you multiply from right to left, already from the start you have something difficult.

Of course, with matrices, it's actually the multiplication itself that gets difficult, not remembering the numbers.

We have names for the two different ways of computing the product:

- backward propagation: compute from left to right;
- forward propagation: compute from right to left.

Clearly we should do backward propagation when $n_k \ll n_0$ and forward propagation when $n_0 \ll n_k$ since we will be working with smaller matrices.

An extreme example is when $n_k = 1$, which is particularly relevant because in optimization we work with real-valued functions $f : \mathbb{R}^n \rightarrow \mathbb{R}$ so $n_0 = n$ and $n_k = 1$.

Consider the case when we have

$$f = f_k \circ f_{k-1} \circ \cdots \circ f_2 \circ f_1$$

where the last function $f_k : \mathbb{R}^{n_{k-1}} \rightarrow \mathbb{R}$ is a real-valued function.

$$f = f_k \circ f_{k-1} \circ \cdots \circ f_2 \circ f_1$$

In deep neural networks, we often have compositions like this, where $f_1, \dots, f_{k-1}$ alternates between affine functions and activation functions and the last f_k is a loss function.

In this case it is standard to express in terms of gradient so

$$\underbrace{\nabla f(\mathbf{x})}_n = \underbrace{Jf_1(\mathbf{x})^\top}_{n \times n_1} \underbrace{Jf_2(f_1(\mathbf{x}))^\top}_{n_1 \times n_2} \cdots \underbrace{Jf_{k-1}(f_{k-2} \circ \cdots \circ f_1(\mathbf{x}))^\top}_{n_{k-2} \times n_{k-1}} \cdot \underbrace{\nabla f_k(f_{k-1} \circ \cdots \circ f_1(\mathbf{x}))}_{n_{k-1}}.$$

Hence if we backpropagate, we start from the right most term, which is a vector of dimension n_{k-1} , then in every step we multiply the vector by a matrix, so we end up with $k - 1$ matrix-vector products.

On the other hand, if we start from the left most term, which is a matrix of dimension $n \times n_1$, then in every step we have to do a matrix-matrix product until the last step when we do a matrix-vector product.

Since matrix-matrix products are much more expensive than matrix-vector products, backpropagation is clearly a better idea when we have to evaluate a gradient of this form.